

Test Suite: Song Music Theory Analysis

Test 1 – Unit Test

What it tests: Step 2 in isolation. Can the agent correctly search Spotify and return the right track ID?

Input prompt:

"What is the Spotify track ID for 'Shape of You' by Ed Sheeran?"

Expected output:

Track ID: 7qiZfU4dY1lW1lzX7mPBI3 Track Name: Shape of You Artist: Ed Sheeran

Pass criteria: Agent calls Spotify search endpoint and returns the correct track ID
7qiZfU4dY1lW1lzX7mPBI3

Test 2 – Workflow Test

What it tests: Full end-to-end multi-step sequence. All 3 APIs called, all fields populated.

Input prompt: *"Give me a full music theory analysis of 'Blinding Lights' by The Weeknd"*

Expected output must include:

- Spotify search returns a valid track ID for Blinding Lights
- ReccoBeats returns audio features including tempo ~171 BPM, low valence (~0.334), high energy (~0.73)
- Key: C# major (key=1, mode=1)
- MusicBrainz returns release year 2019
- Mood label: Angry/Intense (low valence + high energy)
- Output follows the exact structured format from the skill file

Pass criteria: All 3 APIs called AND output follows exact structured format AND all audio features interpreted (not just raw numbers) AND explicit mood label present.

Test 3 — Capability Test

What it tests: Does the agent recognize when and how to apply the skill without being told which tools to use?

Input prompt:

"I'm building an upbeat playlist. Can you analyze 'Happy' by Pharell Williams and tell me if it fits the mood?"

Expected output:

- Agent independently recognizes this requires Spotify + ReccoBeats + MusicBrainz
- Identifies high valence + high energy → Euphoric/Upbeat mood
- Concludes the song fits an upbeat playlist with a justified music theory explanation

Pass criteria: Agent correctly applies the full workflow without being prompted to, identifies the correct mood label, and gives a justified recommendation